

CELL-MEDIATED IMMUNITY

- The immunity developed by cell-mediated response.
- Also called cellular immunity / T-cell immunity.
- Involves several types of cells :—
 - T-lymphocytes
 - Macrophages
 - Natural killer cells
- Does not involve antibodies.
- Major defense mechanism against infections by viruses, fungi.
- Also responsible for: Delayed allergic reactions & rejection of transplanted tissue.

ANTIGEN-PRESENTING CELLS (APCs)

- Induce the release of antigenic materials from invading organisms & later present these materials to helper T-cells.

Types of Antigen-Presenting Cells

- a. **Macrophages (Major APC)** → Phagocytic cells
- b. **Dendritic cells** → Non-phagocytic cells:
 - i. Dendritic cells of spleen (*in blood*)
 - ii. Follicular dendritic cells (*in lymph*)
 - iii. Langerhans dendritic cells in skin (*body surface*)
- c. **B-lymphocytes** :— B-cells ingest foreign bodies via **pinocytosis**.

MHC (MAJOR HISTOCOMPATIBILITY COMPLEX) & HLA (HUMAN LEUKOCYTE ANTIGEN)

- Large molecule present in **short arm of chromosome 6**.
- Made up of **more than 200 genes** (*HLA genes also*).
- **HLA** → Encodes antigen-presenting proteins on cell surface.

MHC is of 2 Types :—

- a. **Class I MHC molecule:**
 - Found on every cell.
 - Responsible for presentation of **endogenous antigens** to helper T-cells.
- b. **Class II MHC molecule:**
 - Found on B-cells, macrophages.
 - Responsible for presenting **exogenous antigens** to helper T-cells.

PRESENTATION OF ANTIGEN

Helper T-cells recognise
antigen



With help of its own
surface receptor protein
(T-cell receptor)



Initiates a complex
interaction between
antigen & T-cell receptor



Same time, macrophages
release IL-1
(Facilitates activation &
proliferation of helper T-
cells)



Proliferated cells enter
circulation



Simultaneously, antigen
activates B-cells also



Development of humoral
immunity

ROLE OF HELPER T-CELLS

- **CD₄ cells enter the circulation** and activate all other T-cells & B-cells.
- **Types :—**
 - a. **Helper-1 (Th₁) cells** → Concerned with cellular immunity:
 - *Secrete:* IL-2, γ -interferon
 - b. **Helper-2 (Th₂) cells** → Concerned with humoral immunity:
 - *Secrete:* IL-4, IL-5
- ↓
- i. Activate B-cells
- ii. Proliferation of plasma cells
- iii. Production of antibodies by plasma cells

ROLE OF CYTOTOXIC T-CELLS (KILLER CELLS)

- Cytotoxic T-cells activated by helper T-cells
- ↓
- Circulate through blood, lymph, lymphatic tissues
- ↓
- **Destroy invading organism by attacking them directly.**

Mechanism of Action:

Receptors on outer membrane bind antigens tightly



Cells enlarge & release substance (lysosomal enzymes)



Substance destroy invading organism

Other Actions:

- a. Also destroy cancer cells.
- b. Destroy body's own tissue (*which are affected by foreign bodies*).

ROLE OF SUPPRESSOR T-CELLS

- Also called regulatory T-cells.
- **Suppress the activities** of killer T-cells / helper T-cells also.
- **Preventing the killer T-cells** from destroying body's own tissues along with invaded organisms.

ROLE OF MEMORY T-CELLS

- T-cells activated by an antigen do not enter circulation but **remain in lymphoid tissue**.
- **When second time organism enters body:**
- ↓
- Memory cells identify & immediately activate T-cells.

Specificity of T-cells :—

Each T-cell activated only by one type of antigen, developing immunity against that antigen. This property is called **specificity of T-cells**.

SCHEMATIC DIAGRAM SHOWING DEVELOPMENT OF

IMMUNITY



